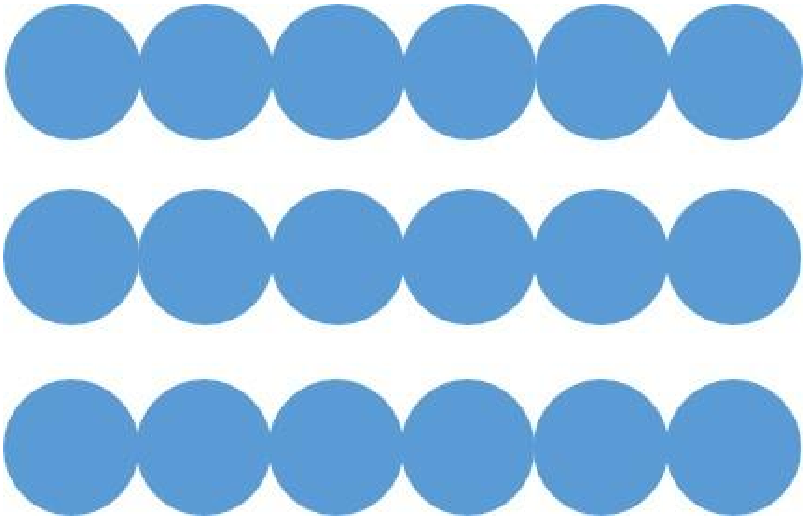
41680 F23 Introduction to advanced materials exam (adjustet for F25)

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:
Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"
There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

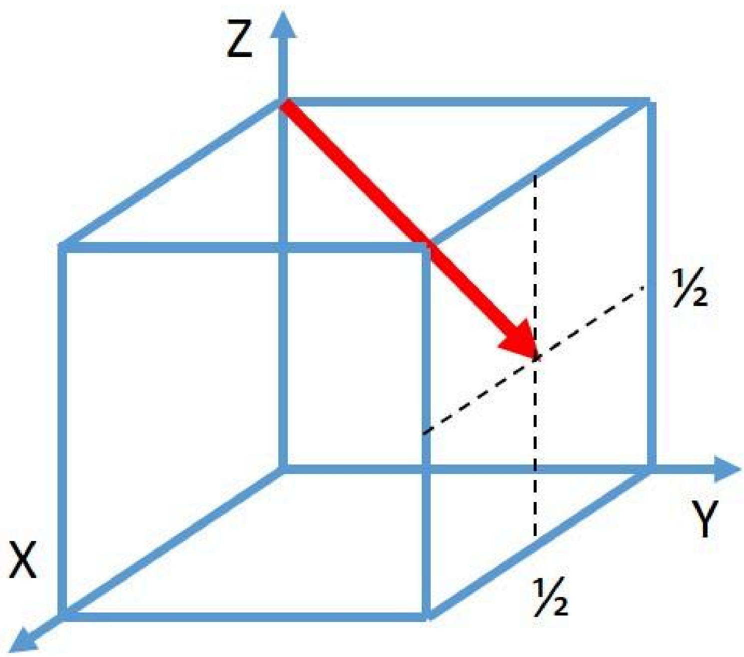
Which crystallographic plane is shown in the figure?



Vælg en svarmulighed

- A {100} plane in a body-centered cubic lattice.
- A {110} plane in a body-centered cubic lattice.
- A {100} plane in a face-centered cubic lattice.
- A {110} plane in a face-centered cubic lattice.
- A {100} plane in a simple cubic lattice.
- A {110} plane in a simple cubic lattice.

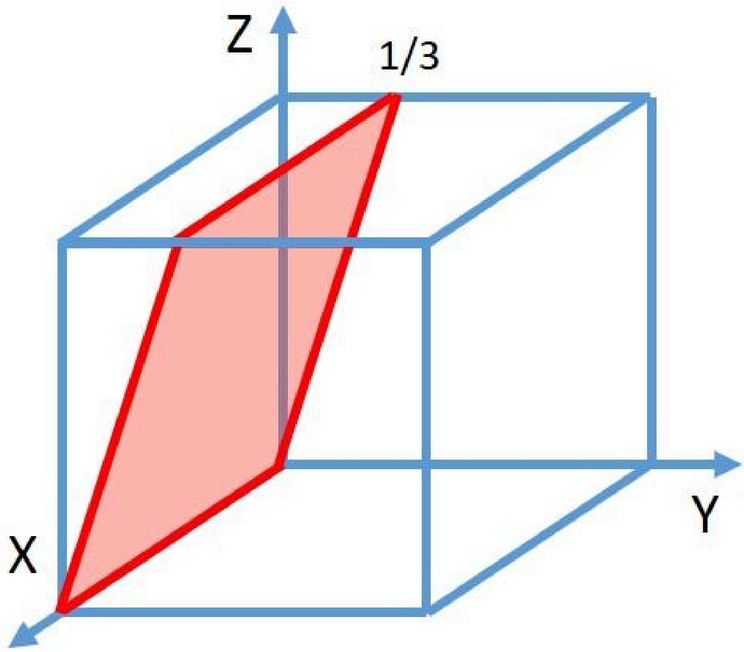
What is the proper notation of the particular crystallographic direction shown in the unit cell in the figure?



Vælg en svarmulighed

- $[12\bar{1}]$
- $[21\bar{2}]$
- $[\bar{1}\bar{2}1]$
- $[\bar{2}\bar{1}2]$

What is the proper notation of the particular crystallographic plane shown in the unit cell in the figure?



Vælg en svarmulighed

- $(03\bar{1})$
- $(01\bar{3})$
- $\{0\bar{1}3\}$
- $\{310\}$

Atoms of the element antimony crystallize under high pressure into a cubic phase with a mass density of 7.48 g/cm^3. The molar mass of antimony is 121.7 g/mol, its atomic radius is 0.150 nm. Which crystal structure has that cubic solid phase?

Vælg en svarmulighed

- ○ A face-centered cubic structure.
- ○ A body-centered cubic structure.
- ○ A simple cubic structure.
- ○ A diamond structure.

The fictive element Melinium (Ml) forms an oxide and a sulfide, both with cubic crystal structure. The table provides some properties of the elements.

Element	Valence	Elektronegativity (Pauling)	Atomic radius / pm	Ionic radius / nm
Ml	+2	1.7	149	95
O	-2	3.5	60	140
S	-2	2.5	106	184

Which statement concerning the compounds MlO and MlS are correct?

Vælg en svarmulighed

- ○ Both compounds have the same crystal structure and their fraction of ionic bonding is at least 50 %.
- ○ Both compounds have the same crystal structure and at least one of the compounds has a fraction of ionic bonding of less than 50 %.
- ○ Both compounds have different crystal structures and their fraction of ionic bonding is at least 50 %.
- ○ Both compounds have different crystal structures and at least one of the compounds has a fraction of ionic bonding of less than 50 %.

Complete miscibility between two metallic elements requires similarity of several material properties. Which of the mentioned material properties will impede complete miscibility in case of a too large difference?

Vælg en svarmulighed

- ◯ Young’s modulus
- ◯ Poisson ratio
- ◯ Electronegativity
- ◯ Electrical resistivity
- ◯ Mass density
- ◯ Molar mass

Where are foreign atoms **not** to be expected, when a pure metal is alloyed with a second element of slightly smaller atomic size?

Vælg en svarmulighed

- On regular lattice sites.
- On interstitial sites.
- At dislocations.
- At grain boundaries.
- At vacancies.

Which point defects have the least effect on strengthening of a metal?

Vælg en svarmulighed

- ○ Vacancies
- ○ Self-interstitial atoms
- ○ Large substitutional atoms
- ○ Small interstitial atoms

By increasing the dislocation density in an aluminium specimen from $1 \cdot 10^{12}$ m^{-2} to $4 \cdot 10^{14}$ m^{-2}, its yield stress increases by 230 MPa. Which additional strengthening is achieved by increasing the dislocation density even further from $4 \cdot 10^{14}$ m^{-2} to $8 \cdot 10^{14}$ m^{-2}?

Vælg en svarmulighed

- ○ About 100 MPa.
- ○ About 230 MPa.
- ○ About 330 MPa.
- ○ About 460 MPa.

A specimen of pure gold is cooled slowly from room temperature to the temperature of liquid nitrogen (-196 °C). How does this affect the defect content?

Vælg en svarmulighed

- The vacancy concentration decreases.
- The dislocation density decreases.
- The grain size decreases.
- The density of grain boundaries decreases.
- Not at all.

Copper is tested in tension. Initially the specimen is loaded to the 0.2 % proof stress. What happens when the specimen is tested further and the stress increases by a factor of 10 % (which still remains below the fracture strength)?

Vælg en svarmulighed

- ◯ The plastic strain increases by 10 %, while the elastic strain remains unchanged.
- ◯ Both strains, the plastic and the elastic strain, increase by 10 %.
- ◯ The plastic strain increases by more than 10 %, while the elastic strain remains unchanged.
- ◯ The elastic strain increases by 10 %, while the plastic strain increases even more.

A copper smith hammers a piece of α-brass (a single phase Cu-Zn alloy with less than 38 wt.% Zn) into a new shape at room temperature. Why does the material become stronger?

Vælg en svarmulighed

- ○ The atoms are squeezed together.
- ○ Cu and Zn atoms are mixed in a better way.
- ○ The grain size becomes reduced.
- ○ More dislocations are formed.

Stiffness and toughness of a metal depend on temperature. When a body-centered cubic metal is cooled to extremely low temperatures, …

Vælg en svarmulighed

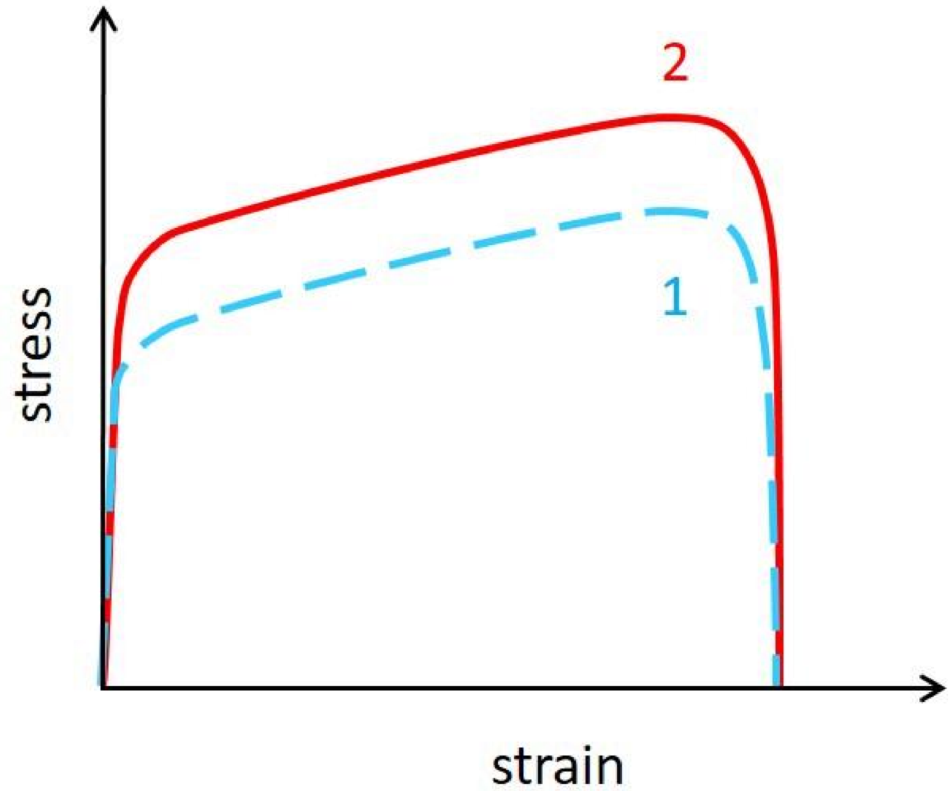
- … both, its stiffness and its toughness, increase.
- … its stiffness increases, but not its toughness.
- … its toughness increases, but not its stiffness.
- … neither its stiffness, nor its toughness increases.

How is crack growth in metals under tension affected by their ability to deform plastically?

Vælg en svarmulighed

- ○ Cracks cannot grow at all when the metal can deform plastically.
- ○ Cracks grow slower when the metal can deform plastically.
- ○ Cracks grow faster when the metal can deform plastically.
- ○ The ability to deform plastically has no effect on crack growth.

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.

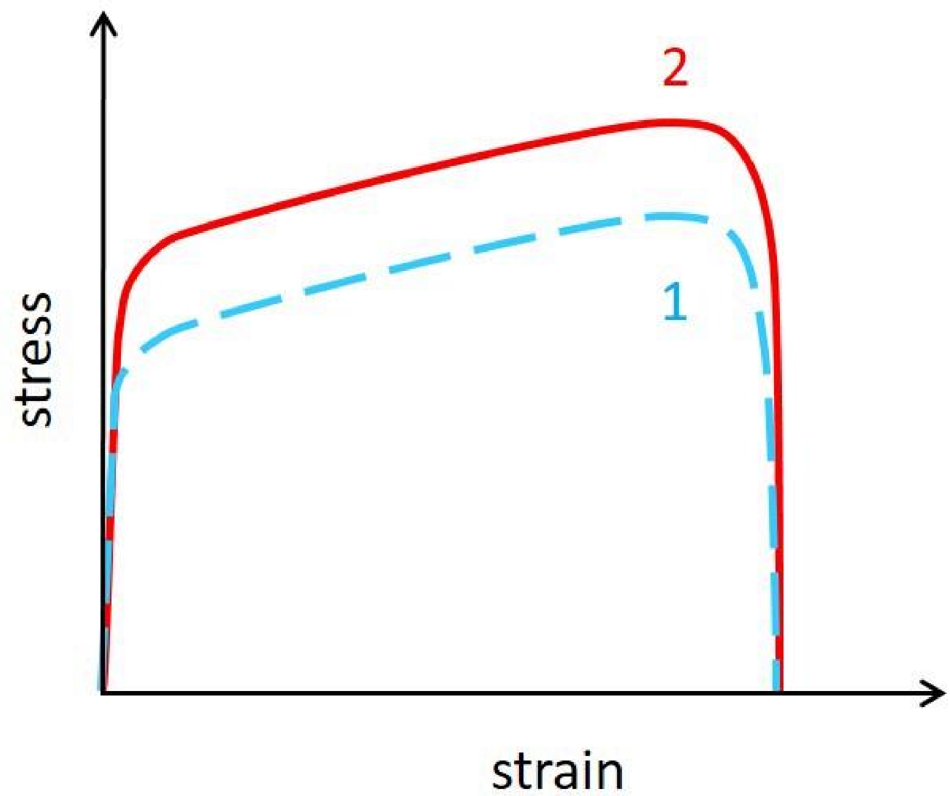


Compare tensile strength and toughness of the two metallic specimens. Which statement is correct?

Vælg en svarmulighed

- ◯ Specimen 1 has both a higher tensile strength and a higher toughness than specimen 2.
- ◯ Specimen 1 has a higher tensile strength, but a lower toughness than specimen 2.
- ◯ Specimen 1 has not a higher tensile strength, but a higher toughness than specimen 2.
- ◯ Specimen 1 has neither a higher tensile strength, nor a higher toughness than specimen 2.

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.

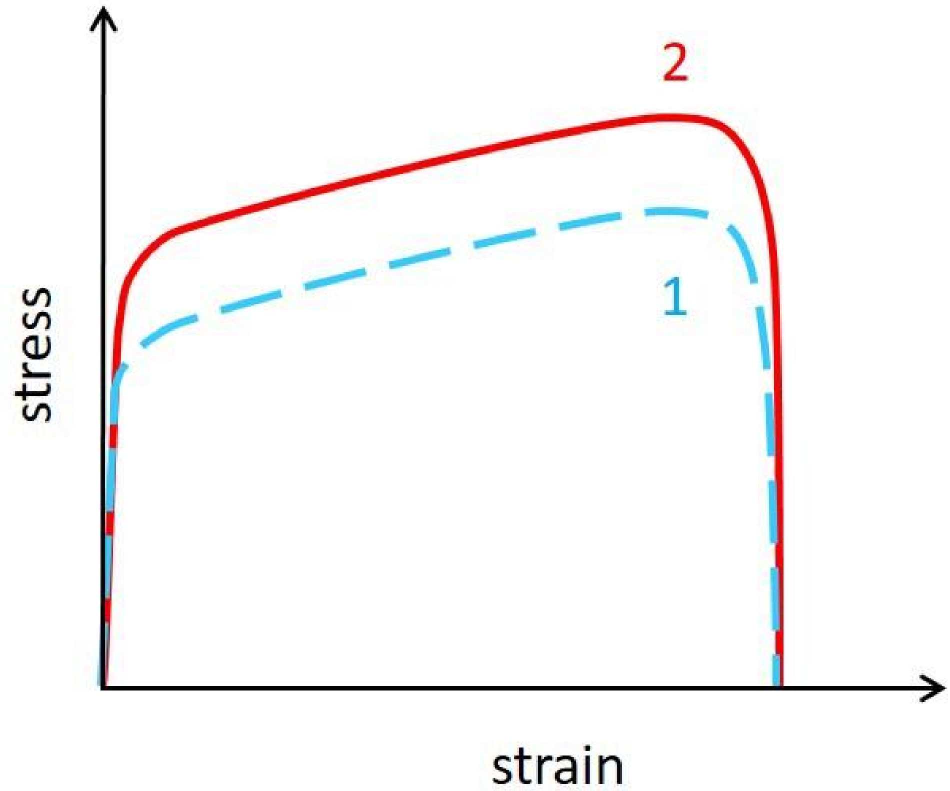


What could be the difference between the two metallic specimens?

Vælg en svarmulighed

- Both specimens may have the same composition, but they have been work-hardened to a different extend.
- Both specimens may have the same composition, but a different grain size.
- Both specimens may have the same composition, but one must have larger grains and must also have been work-hardened to a larger extend than the one with smaller grain size.
- The two specimens must obviously have a different chemical composition.

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.



Which of the two metallic specimens should be selected for a light and stiff pillar of a given length?

Vælg en svarmulighed

- ◯ Specimen 1, if both specimens have the same mass density.
- ◯ Specimen 2, if both specimens have the same mass density.
- ◯ Both specimens are equally suited, if both specimen have the same mass density.
- ◯ The specimen with lower mass density, independent of the flow curve.

A laminate composite is manufactured from several plates made of three different metals (aluminium, copper and lead) such that their volume fraction is the same. The table summarizes mechanical properties of the three metals.

Metal	Young's modulus / GPa	Yield strength / MPa	Tensile strength / MPa	Ductility / %
Al	71.5	128	221	10
Cu	130	130	283	17
Pb	15.4	25.4	36.7	25

Which metal is the first to deform plastically when the laminate is loaded in tension along the plates?

Vælg en svarmulighed

- Al yields first.
- Cu yields first.
- Pb yields first.
- Due to the mechanical restrictions, all metals in the composite must yield simultaneously.

A piece of jewelry is manufactured as layered composite from several thin foils made of two different metals (copper and gold). For economic reasons, the volume fraction of gold is only 15 %. The table summarizes mechanical properties of both metals.

Metal	Young's modulus / GPa	Yield strength / MPa	Tensile strength / MPa	Ductility / %
Cu	130	130	283	17
Au	78.5	184	199	3.5

What is the Young's modulus of the layered composite under compression perpendicular to the foils?

Vælg en svarmulighed

- At least 125 GPa.
- At least 120 GPa, but less than 125 GPa.
- At least 105 GPa, but less than 120 GPa.
- At least 90 GPa, but less than 105 GPa.
- At least 85 GPa, but less than 90 GPa.
- Less than 85 GPa.

High-density polyethylene (HDPE) and low-density polyethylene (LDPE) do not only differ in their density. Which statement is correct when comparing two batches of HDPE and LDPE with same molar mass?

Vælg en svarmulighed

- HDPE has both a higher melting temperature and a higher glass transition temperature than LDPE.
- HDPE has a higher melting temperature, but a lower glass transition temperature than LDPE.
- HDPE has a lower melting temperature, but a higher glass transition temperature than LDPE.
- HDPE has both a lower melting temperature and a lower glass transition temperature than LDPE.

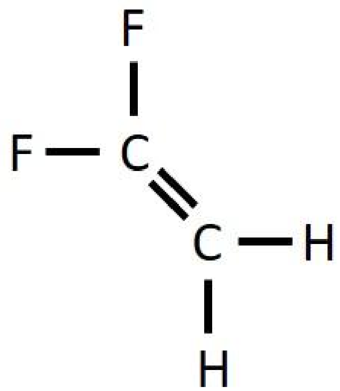
The picture shows plastic beverage cans at room temperature.

What can be concluded about the polymer the cans are made of?

Vælg en svarmulighed

- The polymer must be semi-crystalline and its glass transition temperature must be above room temperature.
- The polymer must be semi-crystalline and its glass transition temperature must be below room temperature.
- The polymer cannot be semi-crystalline and its glass transition temperature must be above room temperature.
- The polymer cannot be semi-crystalline and its glass transition temperature must be below room temperature.

The figure shows the monomer of the polymer polyvinylidene difluoride (PVDF).

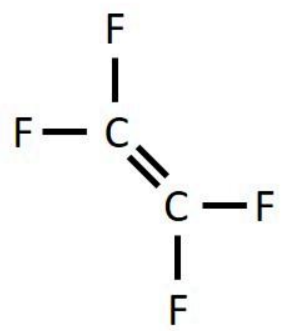


In which condition can PVDF be found?

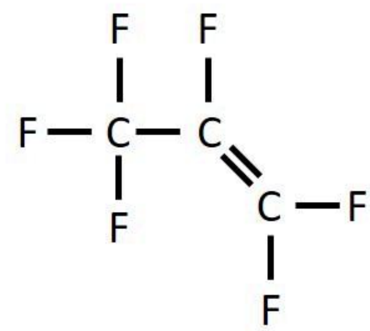
Vælg en svarmulighed

- syndiotactic.
- atactic.
- crystalline.
- semi-crystalline.

Fluorinated ethylene propylene (FEP) is a copolymer of tetrafluoroethylene and hexafluoropropylene, while polytetrafluoroethylene (PFTE) is solely build from tetrafluoroethylene. The figure shows the monomers.



Tetrafluoroethylene



Hexafluoropropylene

Which statement is correct when comparing two batches of FEP and PFTE with the same degree of polymerization?

Vælg en svarmulighed

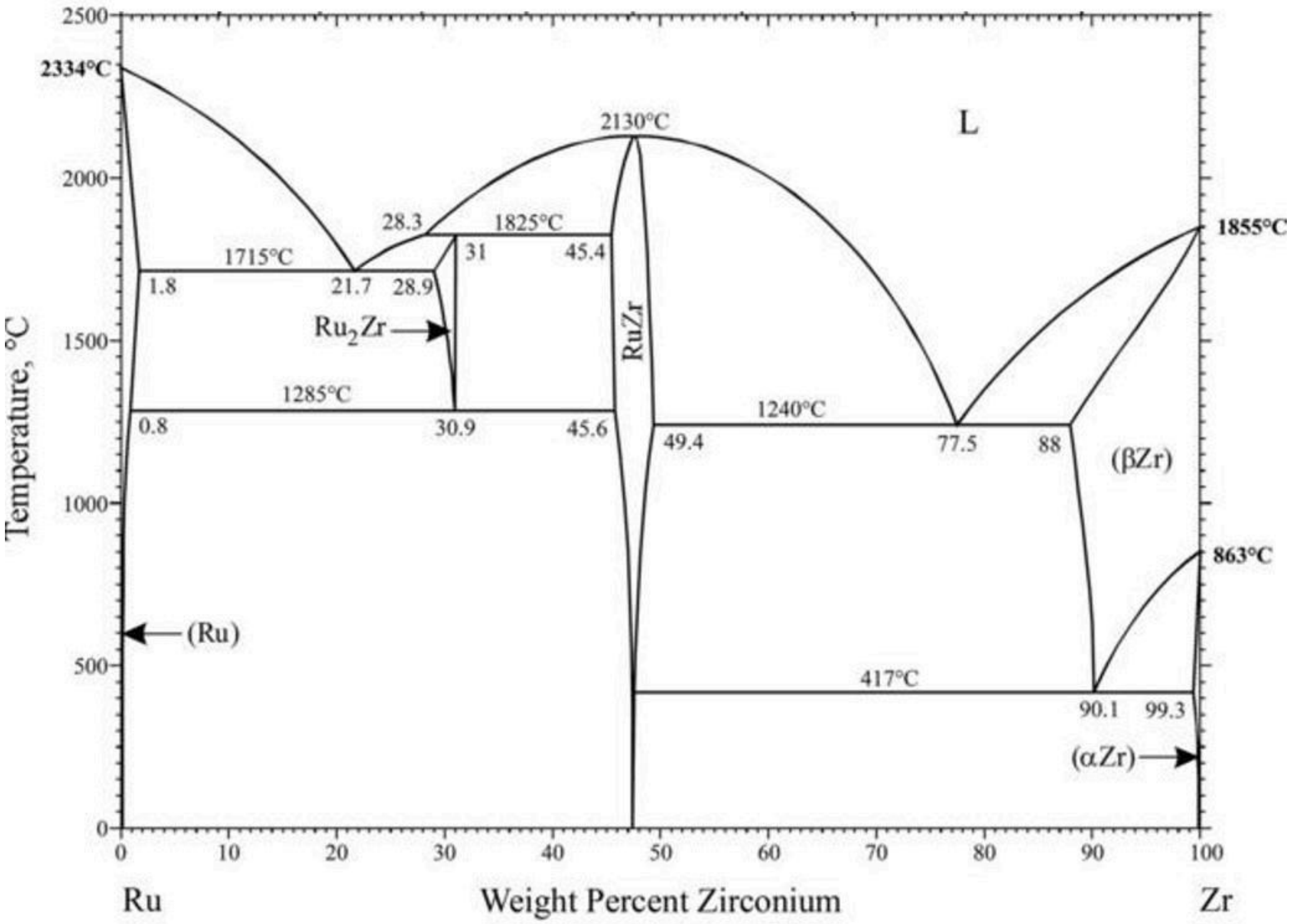
- ◯ FEP has a longer straightened chain length and a higher molar mass than PFTE.
- ◯ FEP has a longer straightened chain length, but not a higher molar mass than PFTE.
- ◯ FEP has not a longer straightened chain length, but a higher molar mass than PFTE.
- ◯ FEP has neither a longer straightened chain length, nor a higher molar mass than PFTE.

Consider a polyethylene molecule without side branches and a molar mass of 5 kg/mol. What is the ratio between the straightened chain length and the average distance between start and end of the chain molecule (carbon-carbon bonds have a length of 0.154 nm and form an angle of 109°)?

Vælg en svarmulighed

- About 18.
- About 15.
- About 13.
- About 11.

The figure shows the phase diagram of the binary alloy system Ru-Zr.

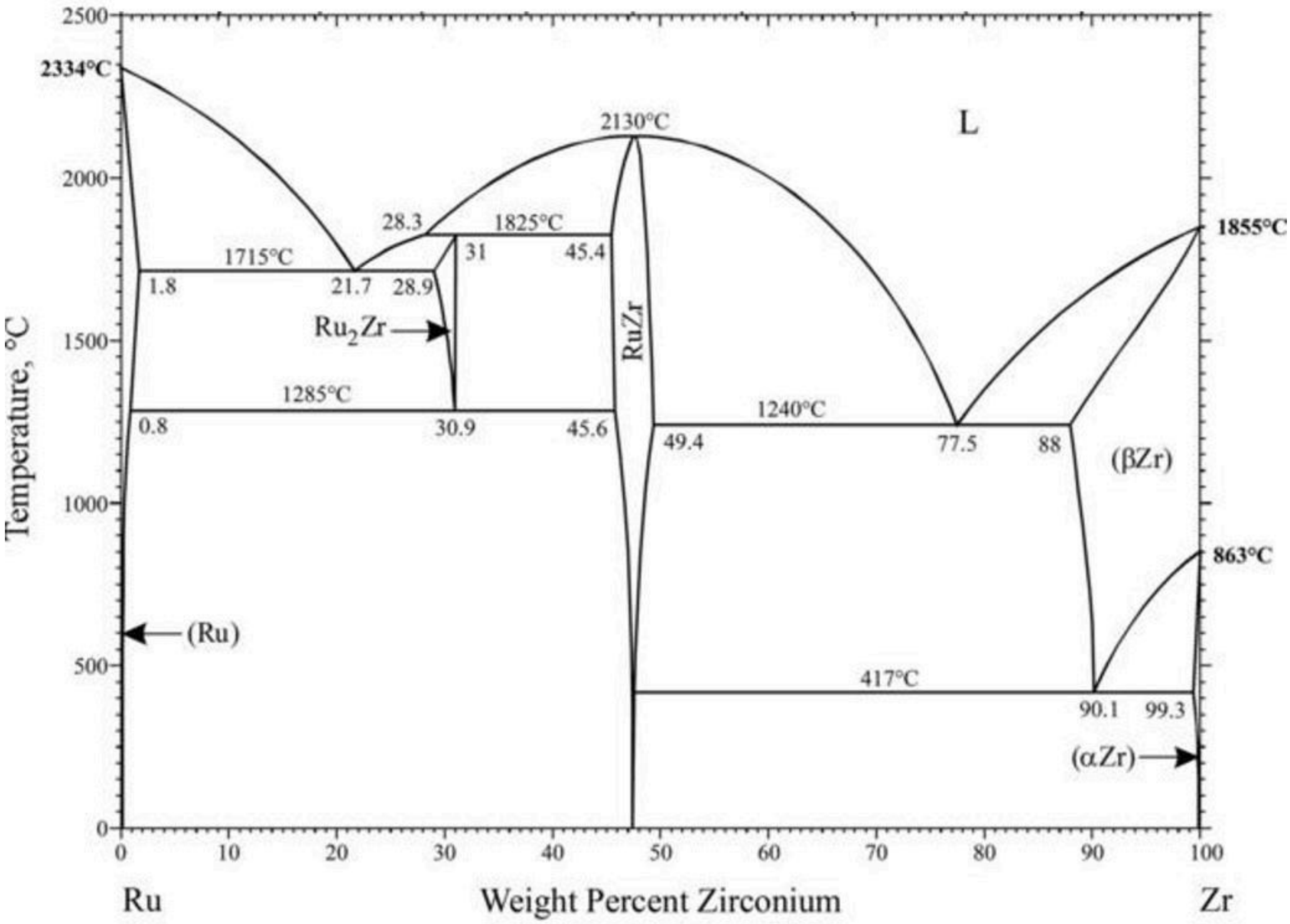


Where does a eutectoid transformation occur in the binary system ruthenium-zirconium?

Vælg en svarmulighed

- ◯ A eutectoid transformation occurs at about 417 °C for 90.1 wt.% Zr.
- ◯ A eutectoid transformation occurs at about 1715 °C for 21.7 wt.% Zr.
- ◯ A eutectoid transformation occurs at about 1825 °C for 31 wt.% Zr.
- ◯ No eutectoid transformation occurs in the system.

The figure shows the phase diagram of the binary alloy system Ru-Zr.

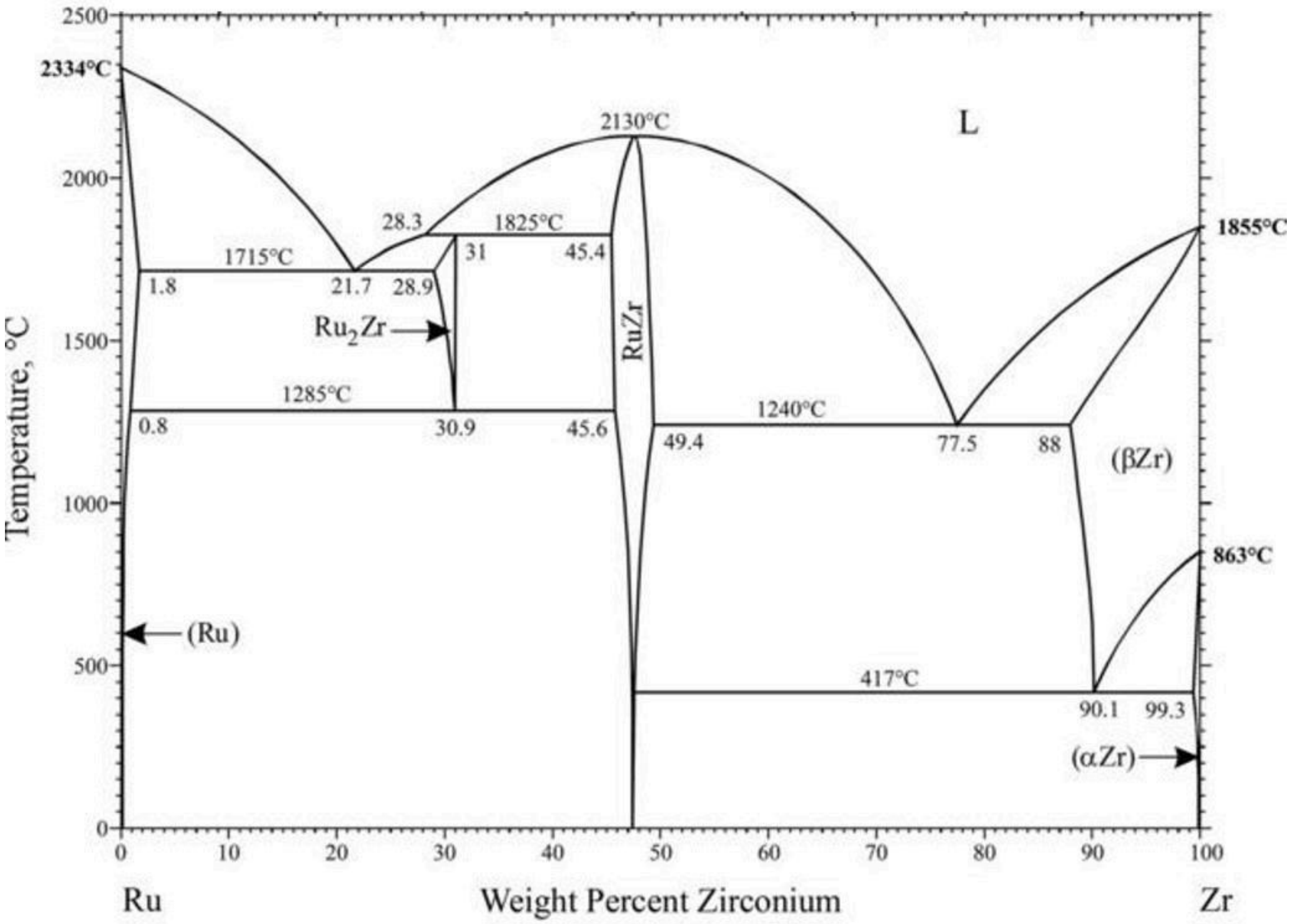


What is the maximum solubility of Ru in solid zirconium?

Vælg en svarmulighed

- 0.7 wt.% Ru
- 12 wt.% Ru
- 88 wt.% Ru
- 99.3 wt.% Ru

The figure shows the phase diagram of the binary alloy system Ru-Zr.

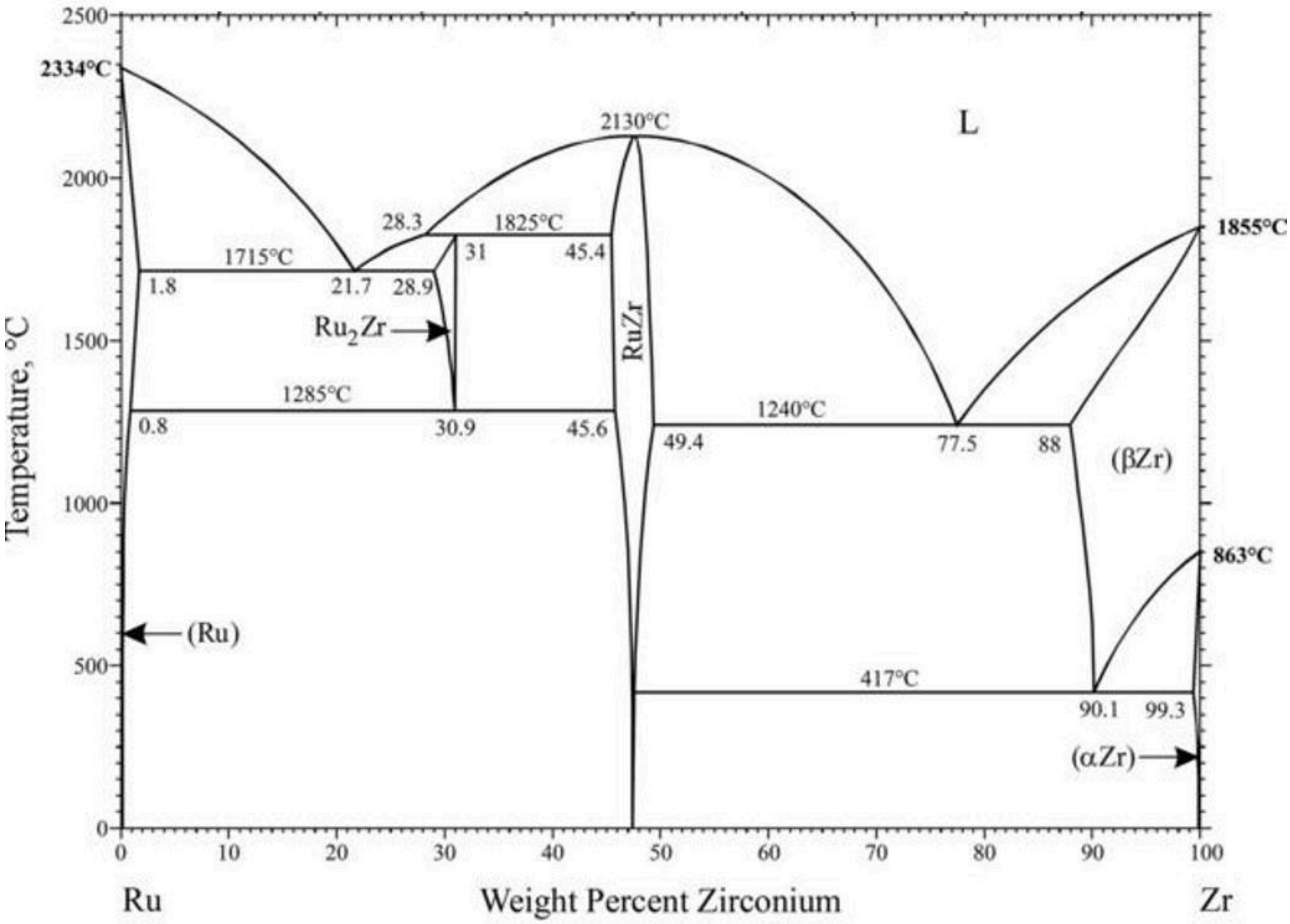


A melt of a ruthenium-zirconium alloy with **38 wt.% Zr** is cooled slowly from 2500 °C to room temperature. Which is the first solid phase formed?

Vælg en svarmulighed

- RuZr
- RuZr + L
- Ru_2Zr
- Ru_2Zr + L
- (Ru)
- (Ru) + L

The figure shows the phase diagram of the binary alloy system Ru-Zr.

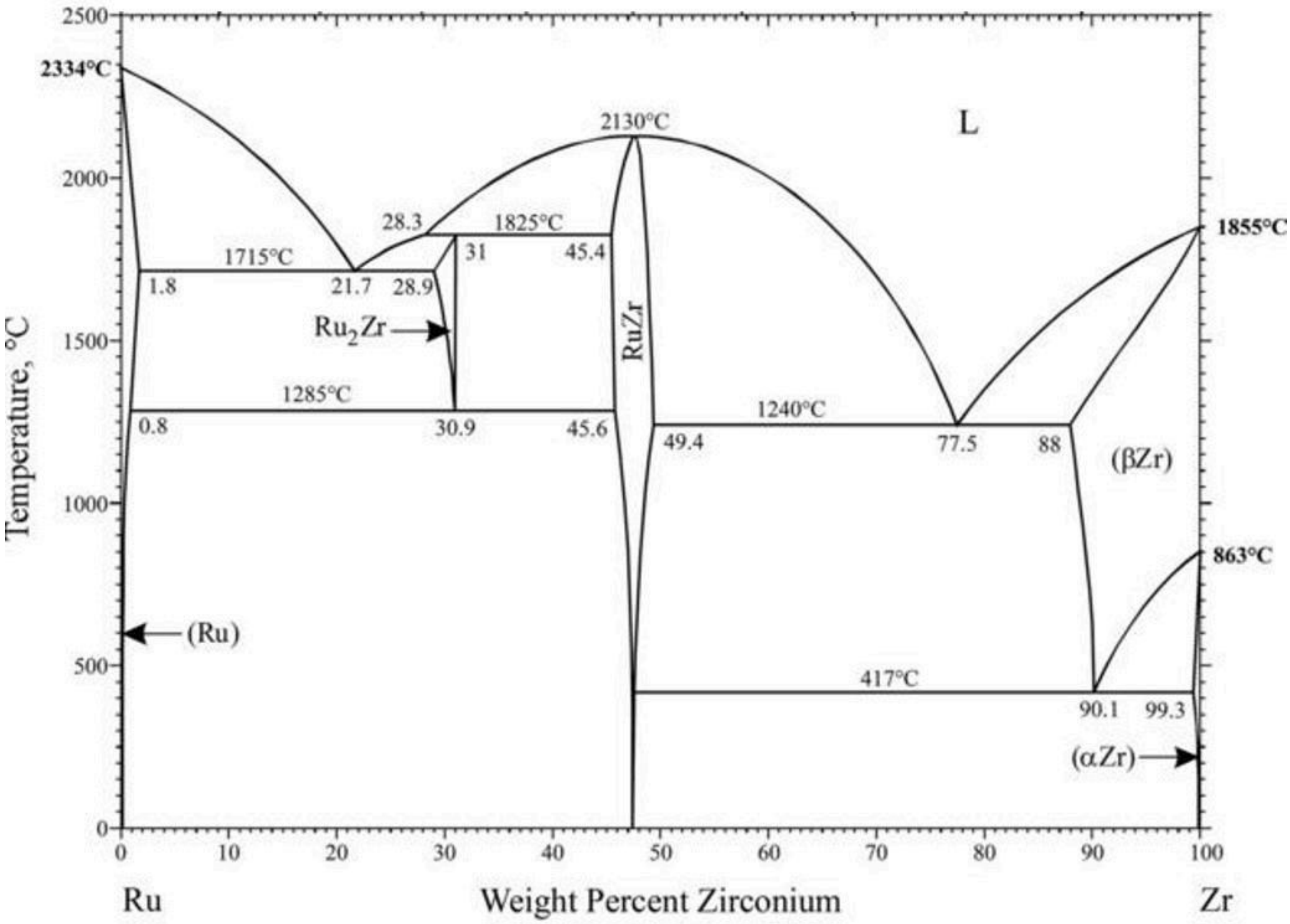


A melt of a ruthenium-zirconium alloy with **38 wt.% Zr** is cooled slowly from 2500 °C to room temperature. What is the lowest Zr content in the melt during solidification?

Vælg en svarmulighed

- 38 wt.% Zr
- 30.9 wt.% Zr
- 28.3 wt.% Zr
- 21.7 wt.% Zr

The figure shows the phase diagram of the binary alloy system Ru-Zr.

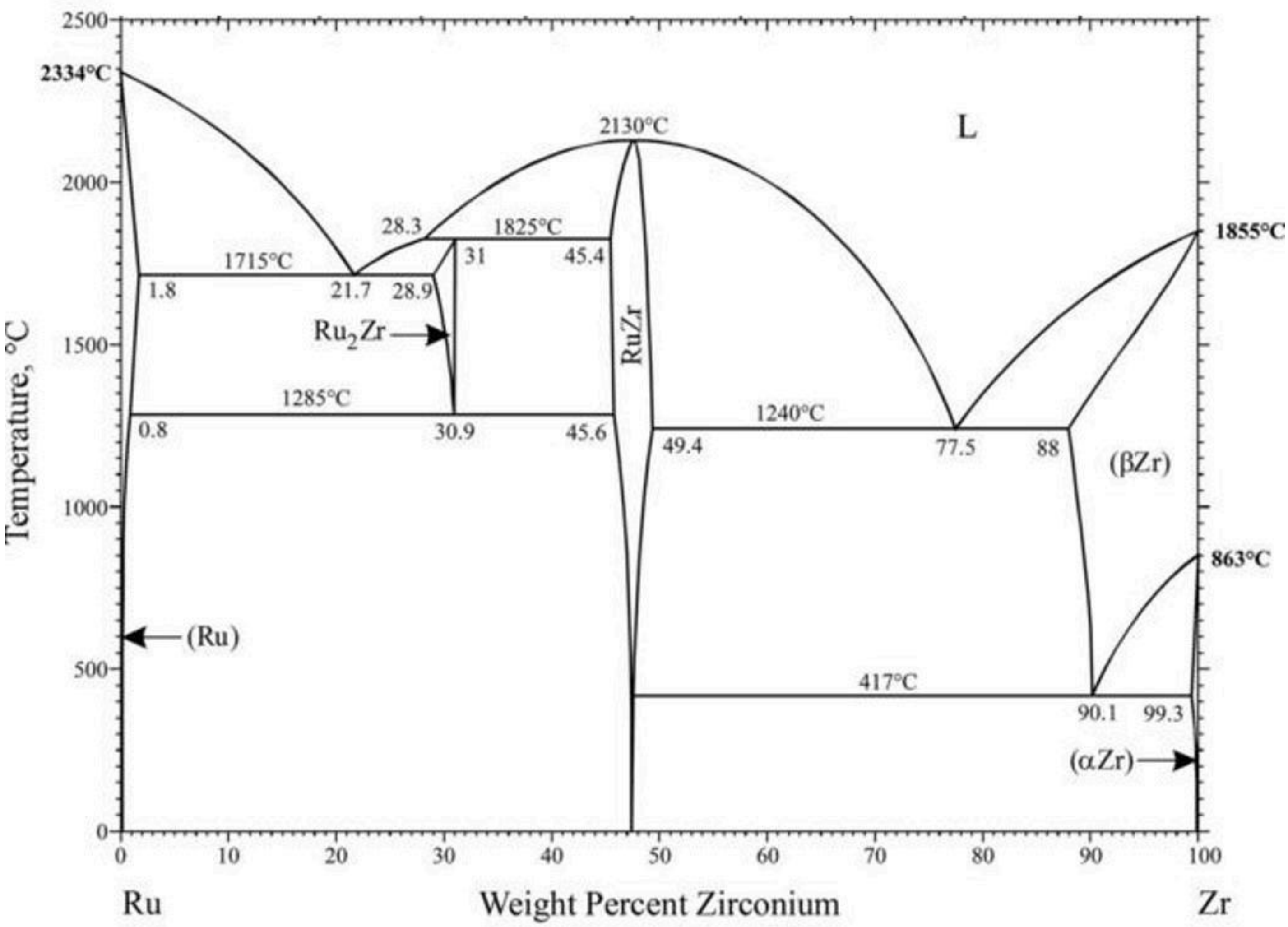


A melt of a ruthenium-zirconium alloy with **77.5 wt.% Zr** is cooled slowly from 2500 °C to room temperature. The microstructure at room temperature consists of two phases. Which phases are present and how are they arranged?

Vælg en svarmulighed

- RuZr and (βZr) as alternating lamellae solely.
- RuZr and (αZr) as alternating lamellae solely.
- RuZr and (αZr) as alternating lamellae with additional RuZr.
- RuZr and (αZr) as alternating lamellae with additional (αZr).

The figure shows the phase diagram of the binary alloy system Ru-Zr.



A melt of a ruthenium-zirconium alloy with **77.5 wt.% Zr** is cooled slowly from 2500 °C to room temperature. What is the mass of RuZr at room temperature for a melt of 6.1 kg?

Vælg en svarmulighed

- About 1.6 kg.
- About 2.6 kg.
- About 3.5 kg.
- About 4.5 kg.

The ceramic material $YBa_2Cu_3O_7$ is a superconductor at low temperatures. At room temperature, $YBa_2Cu_3O_7$ has an electrical conductivity of $5 \cdot 10^5$ S/m. Consider a wire of this material in the shape of a cylinder in an electrical field of 0.1 V/m along its axis. What must the diameter of the wire be, if it carries a current of 4 mA under these conditions at room temperature?

Vælg en svarmulighed

- About 1.6 µm.
- About 8 µm.
- About 320 µm.
- About 1.6 mm.

Consider a hypothetical metal with an electron mobility of 0.002 $m^2/(V\cdot s)$. Moreover, the material has a face-centered cubic structure with a lattice parameter of 0.38 nm, a density of 22.6 g/cm^3, and a molar mass of 192 g/mol. What is the electrical conductivity assuming one conduction electron per atom?

Vælg en svarmulighed

- About $1.4\cdot10^3$ S/m.
- About $6.5\cdot10^6$ S/m.
- About $2.3\cdot10^7$ S/m.
- About $1.4\cdot10^8$ S/m.

Consider the semiconductor Si doped with As. Which of the following statements about the carriers of electricity is correct?

Vælg en svarmulighed

- ○ The carrier densities are approximately constant at all temperatures.
- ○ At low temperatures, the carrier densities are given by the doping concentration, at high temperatures, they become similar to that of undoped Si.
- ○ At low temperatures, there is an equal amount of electrons in the valence band and holes in the conduction band.
- ○ At low temperatures, there is an equal amount of electrons in the conduction band and holes in the valence band

Which of the following statements about phonons is not correct?

Vælg en svarmulighed

- ○ Phonons are related to lattice vibrations in crystalline materials.
- ○ Phonons are a collective motion of atoms, giving rise to waves.
- ○ Phonons transport heat well in crystalline materials.
- ○ Phonons transport electricity well in crystalline materials.

Consider steady state transport of heat from a sphere with a diameter of 2 m and a uniform wall thickness of 15 cm. The wall has a thermal conductivity of 0.2 W/(m·K). How much power is required to be generated inside the sphere, if a constant temperature difference has to be maintained such that the inside is 15 K warmer than the outside of the sphere?

Vælg en svarmulighed

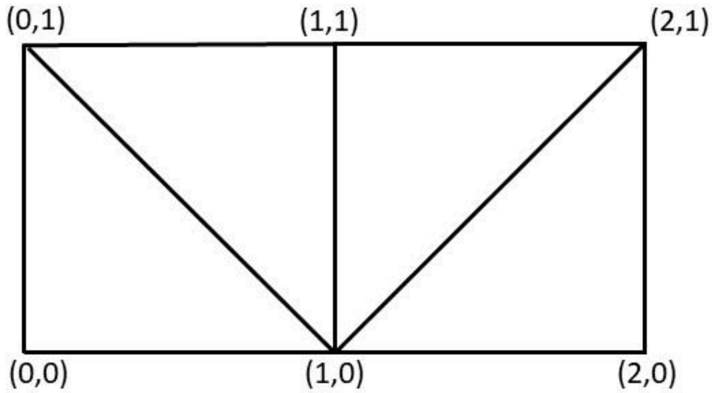
- About 900 kJ/h
- About 1.3 MJ/h
- About 55 MJ/h
- About 130 MJ/h

Consider simulating the temperature distribution in a rod of metal as a 1D heat transfer problem using the heat equation. The rod is 1 m long. The two ends of the rod are defined at x = 0 m (point A) and x = 1 m (point B). In the steady state solution, the temperature in the center of the rod is 30 °C. Which combination of initial temperature (in the entire rod), boundary condition at point A and boundary condition at point B could **not** have produced this result?

Vælg en svarmulighed

- Initial temperature 20° C, point A: u=30° C, point B: $\partial u/\partial n = 0$
- Initial temperature 20° C, point A: $\partial u/\partial n = 0$, point B: $\partial u/\partial n = 0$
- Initial temperature 30° C, point A: u=20° C, point B: u=40° C
- Initial temperature 20° C, point A: u=20° C, point B: u=40° C

Consider an indexed face set representation of the mesh below.



Vertices

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 0 & 0 \\ 1 & 0 \\ 2 & 0 \end{bmatrix}$$

Faces

$$\begin{bmatrix} 1 & 4 & 5 \\ & ? & \\ 2 & 3 & 5 \\ 5 & 3 & 6 \end{bmatrix}$$

What should the second row of the faces matrix be?

Vælg en svarmulighed

- [1 2 3]
- [1 2 5]
- [1 4 6]
- [2 4 5]
- [6 1 3]
- [6 2 5]

Consider the digital image shown.

26	25	18	45	52	38
17	42	20	21	24	21
25	60	51	51	28	06
17	38	74	66	57	11
24	18	14	15	29	04

What is the outcome of applying a 3 x 3 median filter centered at row 2, column 3.

Vælg en svarmulighed

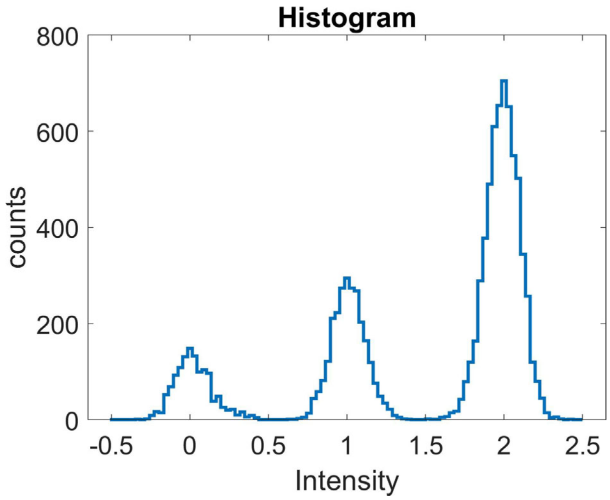
- 20
- 21
- 37
- 38
- 42
- 45

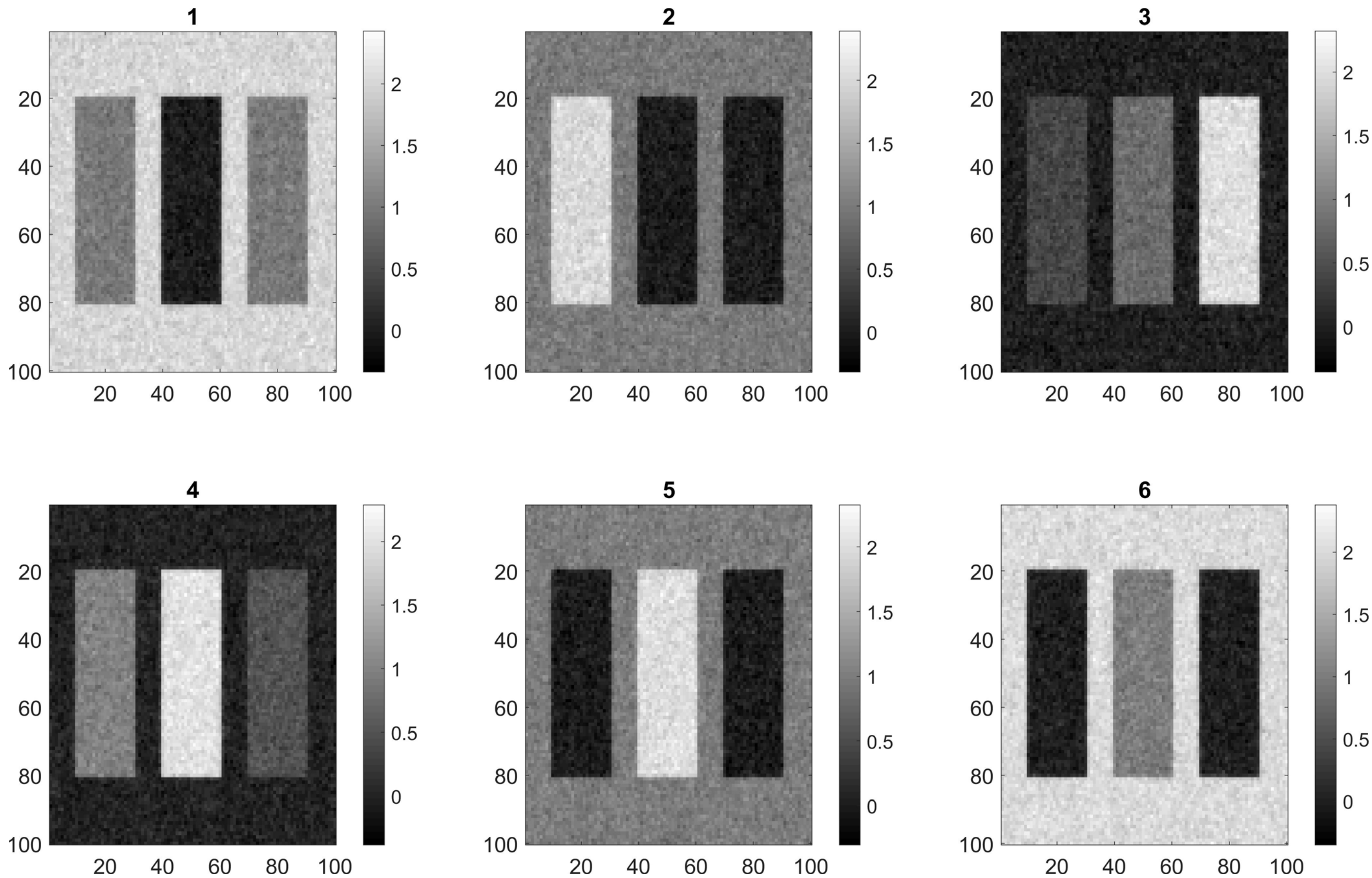
Consider a 3D image of a material consisting of a solid phase and some pores. The image is segmented into solid phase and pores and the volume of the pores is determined to 0.157 mm^3. The 3D image has 300 x 300 x 300 voxels and a voxel size of 20 μm x 20 μm x 20 μm. What is the volume fraction of the pores?

Vælg en svarmulighed

- 0.58
- 0.00058
- 0.73
- 0.00073
- 0.19
- 0.00019

The figure shows the histogram of one of the six images.





Which image does the histogram correspond to?

Vælg en svarmulighed

- 1
- 2
- 3

- ○ 4
- ○ 5
- ○ 6